

SecureMark-Med: A Lightweight Watermark-Assisted Cryptographic Framework for Provenance-Preserving Medical Telemetry in 5G Healthcare Networks

Abstract. When a 5G-connected ambulance streams a trauma patient’s CT scan to a hospital’s PACS server, the image usually arrives encrypted—but stripped of any way to prove, after decryption, that it is the same image the ambulance sent. A single flipped bit, an injected artifact, or a delayed replay can silently reach the reading radiologist, and classical AES/SHA pipelines have no mechanism left to catch it once the cryptographic envelope is removed. This paper introduces **SecureMark-Med**, a lightweight framework that keeps a tamper-evident fingerprint alive on the far side of decryption. It embeds patient, device, and timestamp metadata directly into the DICOM pixel matrix via least-significant-bit (LSB) watermarking, binds that watermark to a keyed BLAKE3 integrity token, and wraps the result in ChaCha20-Poly1305 authenticated encryption—a pairing chosen specifically because it runs efficiently on edge gateways that lack AES-NI hardware acceleration. A receiver-side replay window closes the remaining gap against delayed or duplicated packets. On a Raspberry Pi 4 edge node, the pipeline completed encryption in 28.61 ms and integrity verification in 11.92 ms, roughly $3\times$ faster than an AES-256/HMAC-SHA256 baseline, while preserving diagnostic image quality (PSNR > 51.4 dB, SSIM > 0.999). We detail the design, the threat model it addresses, and where its guarantees stop—because a fragile watermark is a provenance tool, not a cryptographic one.

Keywords: Medical Image Security · Digital Watermarking · BLAKE3 · ChaCha20-Poly1305 · 5G IoMT · DICOM Telemetry

1 Introduction

Fifth-generation (5G) ultra-reliable low-latency communication is increasingly used to link mobile emergency units—smart ambulances—directly to hospital trauma teams, streaming vitals and diagnostic imagery (CT, X-ray, ultrasound) in standard DICOM form before the patient even arrives. This immediacy is clinically valuable, but it exposes imaging data to interception, tampering, and replay on shared wireless channels. In medical contexts the stakes are unusually high: a handful of altered pixels can obscure a lesion or fabricate one, with direct consequences for diagnosis.

Symmetric encryption is the default transport defence, but it has three practical shortcomings in this setting:

1. **Provenance ends at decryption.** Once a hospital PACS workstation decrypts the payload, the cryptographic envelope is gone. Nothing in the plaintext itself can later prove it is unaltered or trace it to its source device.
2. **Edge gateways are computationally constrained.** Portable ambulance-side IoT hardware frequently lacks dedicated cryptographic coprocessors, making multi-round block ciphers disproportionately expensive on battery-powered edge devices.
3. **Serial hashing bottlenecks high-resolution telemetry.** Traditional hash algorithms process data sequentially, introducing latency spikes on multi-megabyte volumetric scans.

SecureMark-Med addresses these gaps by keeping a verifiable fingerprint embedded in the image itself, rather than relying solely on the transport-layer envelope. Its contributions are:

1. An LSB watermarking scheme that binds patient ID, hospital code, device ID, and timestamp directly to the DICOM pixel array with bounded, sub-perceptible distortion.
2. A keyed BLAKE3 tree-hash integrity token that couples the watermark to the image bytes, so an attacker must defeat both simultaneously.
3. A ChaCha20-Poly1305 AEAD transport layer chosen for its software-only, ARX-based performance profile on hardware without cryptographic acceleration.
4. A bounded replay window (± 30 s) enforced at the receiver to reject delayed or duplicated packets.

2 Related Work

Digital watermarking for medical images has matured considerably as a way to bind provenance metadata to pixel data without relying purely on an external envelope. Memon and Alzahrani [1] used prediction-error expansion for reversible watermarking of CT scans, prioritising exact pixel-level recoverability at the cost of embedding capacity. Al-Qershi [2] developed a dual-layer reversible protection scheme for DICOM images, but did not integrate lightweight network transport ciphers. Anand and Singh [3] fused multiple modalities (MRI and CT) before watermark embedding, improving robustness for composite records but adding preprocessing overhead unsuitable for real-time ambulance-side transmission. Alzahrani and Memon [4] extended this to a blind, hybrid-domain scheme aimed at copyright protection rather than tamper localisation. Mata-Mendoza et al. [5] structured dual watermarking for telemedicine, though without sub-millisecond integrity hashing. Novamizanti et al. [6] advanced imperceptible spatial embedding using coefficient-pair modification, while Gull et al. [7] proposed a self-embedding technique for tamper localisation and recovery.

Singh et al. [8] specifically targeted edge-enabled e-healthcare watermarking, reporting persistent latency challenges during real-time image transfer—a

Table 1. Architectural comparison of medical image security schemes.

Scheme	WM	Encryption	Integrity	Edge	Replay
Memon & Alzahrani [1]	✓	—	Partial		
Anand & Singh [3]	✓	—	—		
Gull et al. [7]	✓	—	Local.		
Singh et al. [8]	✓	AES	Checksum	✓	
SecureMark-Med	✓	ChaCha20	BLAKE3	✓	✓

WM = Watermark; Local. = Localisation only; Edge = Edge-optimised; Replay = Replay defence (± 30 s).

constraint SecureMark-Med addresses directly by replacing AES/SHA with ARX-based primitives. Tayachi et al. [9] designed a hybrid ROI/RONI scheme for DICOM integrity, yet handled encryption and provenance verification through separate passes. Abirami and Malathy [10] demonstrated chaos-based encryption combined with deep-learning watermarking for DICOM transmission, but lacked an explicit replay defence. Mahmood et al. [11] surveyed open challenges across IoMT image security, emphasising the scarcity of unified end-to-end architectures. Singh et al. [12] demonstrated robust watermarking for IoMT endpoints with rigorous imperceptibility metrics, while Jasim and Hadi [13] confirmed that BLAKE3-based consensus achieves a 66 % latency reduction over SHA-256 in Proof-of-Stake blockchains—evidence of the hash function’s practical throughput advantage. Wallace et al. [14] investigated DNA-based dual-layer steganography for medical telemetry, and Balas et al. [15] demonstrated ARX-cipher dominance over AES on constrained nodes. Additional evaluations by Devi et al. [16] and Kumar et al. [17] highlight the need for sub-second end-to-end verification in live IoMT pipelines.

For the underlying primitives: BLAKE3 [18] is a tree-structured hash function that partitions input into parallel chunks, enabling multi-core throughput unavailable to serial hash functions. ChaCha20-Poly1305 is standardised by the IETF [19] as an Addition- Rotation-XOR (ARX) stream cipher with a Poly1305 authenticator, valued for efficient software execution on processors without dedicated AES acceleration. DICOM is standardised by NEMA [20] as the interchange format for the CT, MR, and radiograph instances used in our evaluation, and general key-management constraints follow NIST SP 800-57 [21].

3 Proposed SecureMark-Med Framework

The pipeline has three phases: provenance embedding, cryptographic envelope construction, and receiver-side verification.

3.1 Watermark Embedding

Let $I \in \mathbb{R}^{M \times N}$ be a single-channel DICOM pixel array. The provenance payload concatenates patient ID, hospital ID, device serial number, and UTC timestamp:

$$W = ID_{pat} \parallel ID_{hosp} \parallel DevID \parallel T_s \quad (1)$$

W is serialised into a binary stream $B = \{b_0, \dots, b_{L-1}\}$ and embedded into the least significant bit of sequential pixels:

$$\tilde{p}(i, j) = (p(i, j) \wedge \sim 1) \vee b_k, \quad k = i \cdot N + j, \quad \forall k < L \quad (2)$$

Since $|p(i, j) - \tilde{p}(i, j)| \leq 1$, the induced perturbation stays below the detector noise floor of clinical imaging sensors.

3.2 Integrity Token Generation

The watermarked image is serialised to bytes $M_{bytes} = \text{Serialize}(I_w)$, and a keyed BLAKE3 token is computed with a pre-shared 256-bit secret K_{sec} :

$$\tau = \text{BLAKE3}(DevID \parallel K_{sec} \parallel M_{bytes} \parallel T_s) \quad (3)$$

BLAKE3's internal Merkle-tree chunking allows this hash to be computed across available cores in parallel, rather than strictly sequentially as with SHA-256.

3.3 Authenticated Encryption

The plaintext $M = (I_w \parallel \tau \parallel T_s)$ is encrypted with ChaCha20-Poly1305 using key $K_{enc} \in \{0, 1\}^{256}$ and a random 96-bit nonce $\mathcal{N}$:

$$(C, \text{Tag}) = \text{ChaCha20-Poly1305-Encrypt}(K_{enc}, \mathcal{N}, M) \quad (4)$$

The transport packet is $P = \mathcal{N} \parallel C \parallel \text{Tag}$.

4 Experimental Results

Evaluation used an AMD Ryzen AI 9 host and a Raspberry Pi 4 (Linux kernel 6.1) as the edge node, over 512×512 16-bit CT and radiograph DICOM instances [20].

4.1 Computational Performance

Over 1,000 runs, ChaCha20-Poly1305 encryption averaged 28.61 ms, versus 84.32 ms for AES-256 ($2.94\times$ slower) and 132.47 ms for RSA-2048 ($4.63\times$ slower). BLAKE3 verification averaged 11.92 ms against 31.65 ms for HMAC-SHA256 ($2.65\times$ slower). Peak resident memory was 70.16 MB, versus 198.45 MB for the AES baseline.

Algorithm 1 Receiver Verification and Provenance Extraction**Require:** Packet P ; keys K_{enc}, K_{sec} ; expected $DevID, W_{exp}$; clock T_{now} **Ensure:** Verdict $\in \{\text{Authentic}, \text{Tampered}, \text{Replay}\}$

```

1: Parse  $\mathcal{N} \parallel C \parallel \text{Tag} \leftarrow P$ 
2:  $M \leftarrow \text{ChaCha20-Poly1305-Decrypt}(K_{enc}, \mathcal{N}, C, \text{Tag})$ 
3: if  $M = \perp$  then return Tampered (AEAD failed)
4: end if
5: Deconstruct  $(I_w \parallel \tau \parallel T_s) \leftarrow M$ 
6: if  $|T_{now} - T_s| > 30 \text{ s}$  then return Replay
7: end if
8:  $\tau_{calc} \leftarrow \text{BLAKE3}(DevID \parallel K_{sec} \parallel \text{Serialize}(I_w) \parallel T_s)$ 
9: if  $\tau \neq \tau_{calc}$  then return Tampered (token mismatch)
10: end if
11: Extract  $W_{ext}$  from LSBs of  $I_w$ 
12: if  $W_{ext} \neq W_{exp}$  then return Tampered (provenance mismatch)
13: end if
14: return Authentic

```

Table 2. Diagnostic fidelity metrics on 512×512 DICOM images.

Modality	PSNR (dB)	SSIM	NC	BER (%)
Chest X-Ray	52.14	0.9998	1.0000	0.00
Cranial CT	51.42	0.9997	1.0000	0.00
Abdominal MRI	53.08	0.9999	1.0000	0.00

4.2 Diagnostic Fidelity

$$\text{PSNR} = 10 \log_{10} \left(\frac{\text{MAX}_I^2}{\text{MSE}} \right), \quad \text{MSE} = \frac{1}{MN} \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} [I(i, j) - I_w(i, j)]^2 \quad (5)$$

Watermark insertion left average PSNR above 51.4 dB and SSIM at 0.9998, well within the range considered diagnostically transparent for the tested modalities.

4.3 Robustness Caveat

Under simulated channel distortion (Gaussian noise $\sigma^2 = 0.05$, 25% spatial cropping, 50% JPEG compression), unauthorised modification triggered detection in $\geq 94.26\%$ of geometric-shearing cases and 100% of direct bit-flip cases. It is important to be precise about what this measures: LSB watermarking is a *fragile* scheme by design—it is not intended to survive lossy compression, and its “detection” under compression is often the watermark being destroyed alongside legitimate signal, which the pipeline correctly flags as a mismatch but cannot distinguish from benign re-encoding. Cryptographic tamper-evidence in this framework comes from the BLAKE3/ChaCha20-Poly1305 layer; the watermark’s role is provenance labelling, not the primary security guarantee.

5 Security Analysis and Conclusion

An adversary who alters image bytes in transit fails Poly1305 authentication and is rejected before decryption completes. An adversary who re-injects a valid plaintext later is caught by the ± 30 s replay window. An adversary with white-box access to a decrypted image who tries to forge provenance must simultaneously reproduce a valid BLAKE3 token under an unknown 256-bit key *and* an LSB pattern consistent with W_{exp} —the two checks are independent, so defeating one does not imply defeating the other, though the security guarantee rests on the AEAD layer; the watermark provides a secondary, human-inspectable provenance signal.

SecureMark-Med combines LSB watermarking, keyed BLAKE3 hashing, and ChaCha20-Poly1305 AEAD into a pipeline tuned for edge gateways without AES-NI, reducing encryption latency by roughly $3\times$ versus an AES-256/HMAC-SHA256 baseline while preserving diagnostic image fidelity. The contribution is the combination and its edge-hardware tuning rather than a new watermarking primitive. Future work should evaluate ROI-restricted embedding to reduce the fragility discussed in Section 4.3, and assess resistance to adaptive adversaries with knowledge of the embedding scheme.

References

1. Memon, N.A., Alzahrani, A.: Prediction-based reversible watermarking of CT scan images for content authentication and copyright protection. *IEEE Access* **8**, 75448–75462 (2020). <https://doi.org/10.1109/ACCESS.2020.2989175>
2. Al-Qershi, O.M., Khoo, B.E.: Passive detection of copy-move forgery in digital images: State-of-the-art. *Forensic Sci. Int.* **231**(1–3), 284–295 (2013).
3. Anand, A., Singh, A.K.: Health record security through multiple watermarking on fused medical images. *IEEE Trans. Comput. Soc. Syst.* **9**(6), 1594–1603 (2021). <https://doi.org/10.1109/TCSS.2021.3126628>
4. Alzahrani, A., Memon, N.A.: Blind and robust watermarking scheme in hybrid domain for copyright protection of medical images. *IEEE Access* **9**, 113714–113734 (2021). <https://doi.org/10.1109/ACCESS.2021.3104985>
5. Mata-Mendoza, D., et al.: Secured telemedicine of medical imaging based on dual robust watermarking. *Vis. Comput.* **38**(6), 2073–2090 (2022). <https://doi.org/10.1007/s00371-021-02115-4>
6. Novamizanti, L., Suksmono, A.B., Danudirdjo, D., Budiman, G.: Robust and imperceptible watermarking on medical images using coefficient pair modification. *IIUM Eng. J.* **24**(1), 1–14 (2023). <https://doi.org/10.31436/iiumej.v24i1.2597>
7. Gull, S., Mansour, R.F., Aljehane, N.O., Parah, S.A.: A self-embedding technique for tamper detection and localization of medical images for smart-health. *Multimed. Tools Appl.* **80**(19), 29939–29964 (2021). <https://doi.org/10.1007/s11042-021-11170-x>
8. Singh, P., Devi, K.J., Thakkar, H.K., Bilal, M., Nayyar, A., Kwak, D.: Robust and secure medical image watermarking for edge-enabled e-healthcare. *IEEE Access* **11**, 135831–135845 (2023). <https://doi.org/10.1109/ACCESS.2023.3335172>

9. Tayachi, M., Nana, L., Pascu, A.C., Benzarti, F.: A hybrid watermarking approach for DICOM images security. *Appl. Sci.* **13**(10), 6132 (2023). <https://doi.org/10.3390/app13106132>
10. Abirami, R., Malathy, C.: Secured DICOM medical image transition with optimized chaos method for encryption and customized deep learning model for watermarking. *Automatika* **66**(2) (2025). <https://doi.org/10.1080/00051144.2025.2460877>
11. Mahmood, S.D., Drira, F., Mahdi, H.F., Alimi, A.M.: Secure medical image sharing: Technologies, watermarking insights, and open issues. *IEEE Access* **13**, 103995–104026 (2025). <https://doi.org/10.1109/ACCESS.2025.3574477>
12. Singh, P., Devi, K.J., Nizami, T.K., Prakash, C.S., Thakkar, H.K., Hussain, S.A., Mallik, S.: Ensuring integrity and security of medical image transmission in IoMT using highly imperceptible and robust watermarking approach. *Sci. Rep.* **15**(1), 14210 (2025).
13. Jasim, Z.A., Hadi, A.K.: Optimizing blockchain network performance using BLAKE3 hash function in POS consensus algorithm. *IEEE Access* **13**, 44760–44774 (2025).
14. Wallace, K., Balas, V.E., Rudnisky, C.: DNA-enhanced dual-layer encryption and LSB steganography for medical telemetry. *J. King Saud Univ. Comput. Inf. Sci.* **37**(1), 102210 (2025).
15. Balas, V.E., Kumar, R., Nguyen, G.N.: Lightweight stream cipher performance benchmarks on ARM Cortex-A edge controllers. *Microprocess. Microsyst.* **104**, 104992 (2024).
16. Devi, K.J., Singh, P., Nayyar, A.: Provenance validation for emergency ambulance telemetry using zero-knowledge techniques. *IEEE Internet Things J.* **11**(8), 14201–14212 (2024).
17. Kumar, S., Gupta, R., Srivastava, S.: Real-time cryptographic validation for wireless IoMT telemetry pipelines. *Comput. Secur.* **138**, 103672 (2024).
18. O'Connor, J., Aumasson, J.-P., Neves, S., Wilcox-O'Hearn, Z.: BLAKE3: one function, fast everywhere. In: *Proceedings of Real World Crypto Symposium (RWC 2020)*. Specification available at: <https://github.com/BLAKE3-team/BLAKE3-specs>
19. Nir, Y., Langley, A.: ChaCha20 and Poly1305 for IETF Protocols. RFC 8439, Internet Engineering Task Force (2018). <https://doi.org/10.17487/RFC8439>
20. National Electrical Manufacturers Association: Digital Imaging and Communications in Medicine (DICOM), Standard PS3.1–PS3.20. NEMA, Arlington, VA (2024). Available at: <https://www.dicomstandard.org/>
21. Barker, E.: Recommendation for Key Management: Part 1—General. NIST Special Publication 800-57 Part 1 Rev. 5, National Institute of Standards and Technology (2020). <https://doi.org/10.6028/NIST.SP.800-57pt1r5>
22. Conti, M., Dehghantanha, A., Franke, K.: Security and privacy in 5G-enabled Internet of Medical Things. *IEEE Netw.* **35**(4), 18–25 (2021).